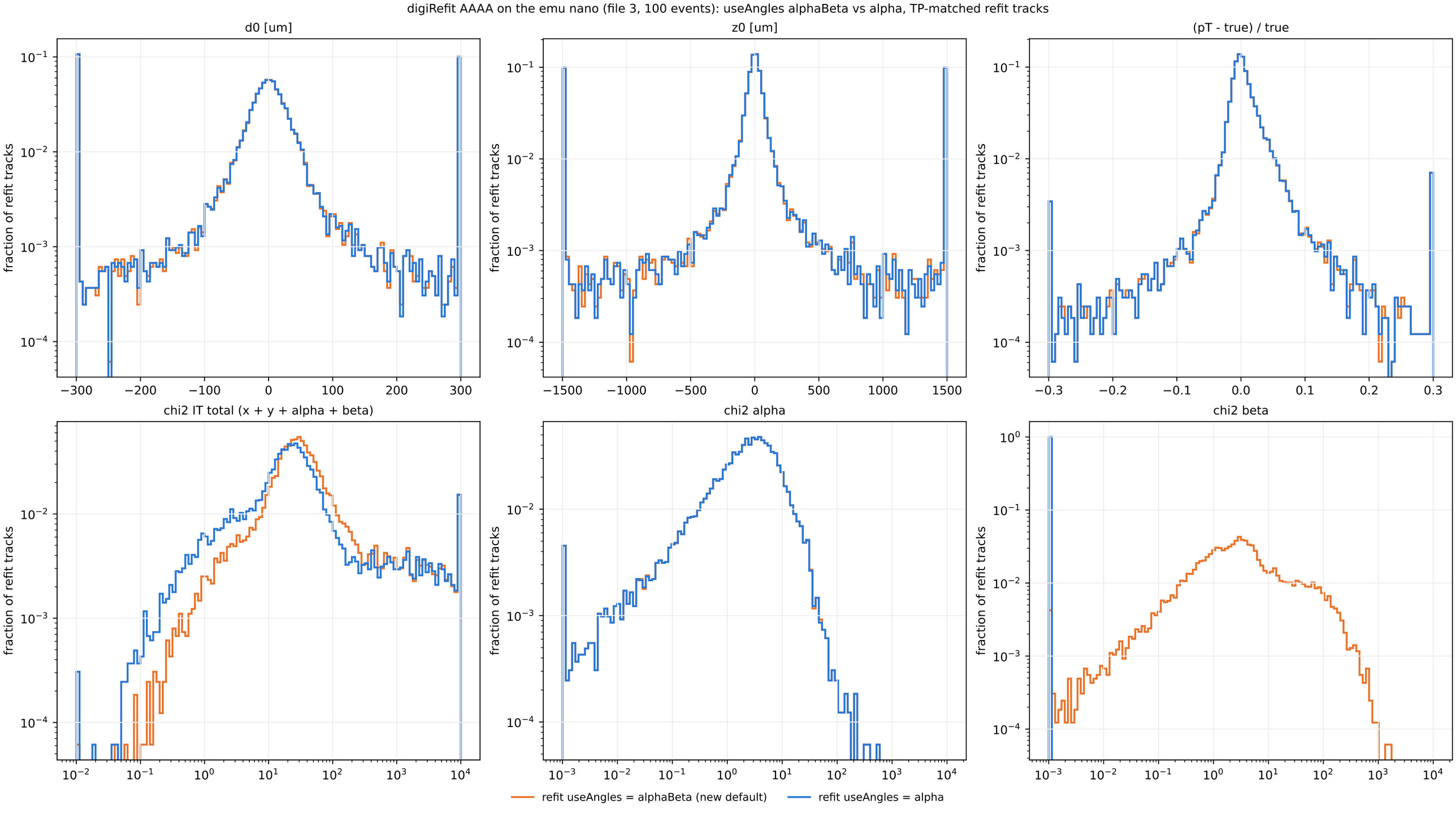
digiRefit AAAA on the emu nano (file 3, 100 events): useAngles alphaBeta vs alpha, TP-matched refit tracks
d0 [um]
z0 [um]
(pT - true) / true
chi2 IT total (x + y + alpha + beta)
chi2 alpha
chi2 beta
fraction of refit tracks
refit useAngles = alphaBeta (new default)
refit useAngles = alpha

Refit residual widths vs the TP (q68 half-width, core RMS, median) and chi2 summaries

refit	quantity	tp_pt	n	q68_halfwidth	core_rms	median	mean	q90	q99
alphaBeta (new default)	d0 [um]	[0, inf)	16303	60.36	25.87	-0.01554	nan	nan	nan
alphaBeta (new default)	d0 [um]	[2, 5)	13309	56.99	27.55	0.03845	nan	nan	nan
alphaBeta (new default)	d0 [um]	[5, inf)	1666	25.25	11.94	0.3917	nan	nan	nan
alphaBeta (new default)	z0 [um]	[0, inf)	16303	158.2	45.89	0.2718	nan	nan	nan
alphaBeta (new default)	z0 [um]	[2, 5)	13309	142.9	48.98	0.2766	nan	nan	nan
alphaBeta (new default)	z0 [um]	[5, inf)	1666	57.24	28.03	0.4715	nan	nan	nan
alphaBeta (new default)	pT (reco - true) / true	[0, inf)	16303	0.0196	0.01256	0.001387	nan	nan	nan
alphaBeta (new default)	pT (reco - true) / true	[2, 5)	13309	0.0183	0.01221	0.0005543	nan	nan	nan
alphaBeta (new default)	pT (reco - true) / true	[5, inf)	1666	0.01851	0.01178	0.001866	nan	nan	nan
alphaBeta (new default)	chi2 IT total	all	16303	nan	nan	30.38	620.6	428.5	1.422e+04
alphaBeta (new default)	chi2 alpha	all	16303	nan	nan	2.477	4.486	9.686	30.62
alphaBeta (new default)	chi2 beta	all	16303	nan	nan	2.711	16.61	40.84	228.3
alpha	d0 [um]	[0, inf)	16303	60.44	25.97	0.0107	nan	nan	nan
alpha	d0 [um]	[2, 5)	13309	57.15	27.4	0.05092	nan	nan	nan
alpha	d0 [um]	[5, inf)	1666	25.25	11.94	0.4102	nan	nan	nan
alpha	z0 [um]	[0, inf)	16303	158.4	45.91	0.2337	nan	nan	nan
alpha	z0 [um]	[2, 5)	13309	143.3	49.12	0.205	nan	nan	nan
alpha	z0 [um]	[5, inf)	1666	57.02	28.24	0.487	nan	nan	nan
alpha	pT (reco - true) / true	[0, inf)	16303	0.01961	0.01259	0.001385	nan	nan	nan
alpha	pT (reco - true) / true	[2, 5)	13309	0.0183	0.01223	0.0005573	nan	nan	nan
alpha	pT (reco - true) / true	[5, inf)	1666	0.01852	0.01178	0.001877	nan	nan	nan
alpha	chi2 IT total	all	16303	nan	nan	22.2	604.5	356.6	1.402e+04
alpha	chi2 alpha	all	16303	nan	nan	2.477	4.488	9.688	30.6
alpha	chi2 beta	all	16303	nan	nan	0	0	0	0

Accepted hits: the |cot| <= 12 cap rule (refit hasAlpha/hasBeta vs the cluster's reported cot) and selected-hit purity (selHitClass)

refit	accepted_hits	hasAlpha_rule_violations	hasBeta_rule_violations	alpha_dropped_by_cap	beta_dropped_by_cap	purity_own_TP	frac_other_TP	frac_no_TP
alphaBeta (new default)	46521	0	0	8	37	0.8544	0.13	0.01567
alpha	46519	0	0	8	37	0.8543	0.13	0.01567